

1. Administrative & Study Identification

Full Study Title: A Study to Investigate the Pharmacokinetics, Safety, Tolerability, and Efficacy of AZD0780 with Ezetimibe Combinations in Healthy Adults with Elevated LDL-C.
Short Title/Acronym: AZD0780 with Ezetimibe Combinations
Protocol Number: D7960C00017
NCT Number: NCT06742853
Posting ID: 869235349604422440
Sponsor / Company: AstraZeneca
Investigational Product: AZD0780 (in combination with Ezetimibe, Ezetimibe/Rosuvastatin, or Ezetimibe/Bempedoic Acid, or Placebo)
Phase of Development: Phase 1
Trial Status: Completed
Medical Monitor: AstraZeneca Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the effect of AZD0780 on LDL-C levels versus placebo, and assess the safety and tolerability of AZD0780 when dosed with ezetimibe or ezetimibe and rosuvastatin or ezetimibe and bempedoic acid.
Secondary Objectives: To evaluate the pharmacokinetic (PK) profile of various AZD0780 combination regimens.

2.2 Background & Rationale To address the need for intensified lipid-lowering therapies for patients who do not achieve their LDL-C targets. This study investigates how the novel treatment AZD0780 functions when combined with standard cholesterol-lowering drugs in healthy individuals with elevated LDL-C levels.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional
Control Type: Placebo-controlled
Blinding: Single-blind
Randomization: Randomized, Parallel Assignment

3.2 Planned Enrollment & Duration Target \$N\$: 81 healthy adult participants
Number of Sites: Multiple clinical research units
Status: Completed

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Signed the informed consent form before any study-related procedure. 2. All females must have a negative serum pregnancy test at the Screening Visit and on admission to the Clinical Unit. 3. Females of non-childbearing potential must be confirmed at the Screening Visit by fulfilling one of the following criteria: postmenopausal or surgically sterilized females. 4. Have a Body Mass Index (BMI) ≥ 18 kg/m² and weigh at least 50 kg. 5. Sexually active fertile male participants with partners of childbearing potential must adhere to the contraception methods. 6. Fasting LDL-C ≥ 100 mg/dL but < 190 mg/dL (≥ 2.6 mmol/L but < 4.9 mmol/L for London EPCU) at the Screening Visit. 7. Fasting triglycerides < 400 mg/dL (or < 10.3 mmol/L for London EPCU) at the Screening Visit.

4.2 Exclusion Criteria 1. History of any clinically important disease or disorder. 2. History or presence of gastrointestinal, hepatic, or renal disease or any other condition known to interfere with absorption, distribution, metabolism, or excretion of drugs. 3. Any clinically important illness, medical/surgical procedure, or trauma within 4 weeks of the first administration of study intervention. 4. Any laboratory values with specific deviations in alanine transaminase (ALT), aspartate transaminase (AST), total bilirubin (TBL), estimated glomerular filtration rate (eGFR), or hemoglobin at the Screening Visit or on Admission. 5. Any clinically important abnormalities in clinical chemistry, hematology, or urinalysis results other than those described under exclusion criterion number 4, at Screening and/or Admission to the Clinical Unit. 6. Any positive result on Screening for serum HBsAg, hepatitis B core antibody, or human immunodeficiency virus (HIV). 7. History of severe allergy/hypersensitivity or ongoing clinically important allergy/hypersensitivity, as judged by the investigator, or history of hypersensitivity to drugs with a similar chemical structure or class to AZD0780, ezetimibe, rosuvastatin, and bempedoic acid. 8. Treatment with any lipid-lowering therapy or AZD0780 within the 3 months prior to Screening. 9. Treatment with drugs for reduction or inhibition of Proprotein convertase subtilisin/kexin type 9 (PCSK9) within the last 12 months prior to Screening (approved or investigational and apart from AZD0780).

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Participants receive AZD0780 or Placebo added to standard doses of ezetimibe, ezetimibe/rosuvastatin, or ezetimibe/bempedoic acid. **Route of Administration:** Oral **Duration of Treatment:** 28 days following a 28-day Run-in Period.

5.2 Concomitant & Prohibited Therapy Permitted: Standard allowed medications as cleared by the investigator.

Prohibited: Use of other lipid-lowering therapies, PCSK9 inhibitors, or any investigational agents prior to or during the study duration.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Up to Day -28	Informed Consent, Medical History, Fasting Lipid Profile, Safety Labs
Run-in Period	Day -28 to 0	Baseline assessments, Stabilization on background combination therapy
Treatment Period	Day 1 to Day 28	Randomization, First Dose of AZD0780/Placebo, PK Sampling, Safety Checks
Follow-up Visit 1	Week 5 (One week post-dose)	Safety assessment, PK/PD Evaluation
Follow-up Visit 2	Week 6 (Two weeks post-dose)	Final Follow-up, Safety Evaluation, Study Conclusion

7. Outcome Measures (Endpoints)

7.1 Primary Outcomes

- Change from baseline in direct Low-density Lipoprotein Cholesterol (LDL-C):** To evaluate the effect of AZD0780 on LDL-C levels versus placebo when dosed with ezetimibe or ezetimibe and rosuvastatin or ezetimibe and bempedoic acid.
- Number of participants with adverse events:** The safety and tolerability of AZD0780 when dosed with ezetimibe or ezetimibe and rosuvastatin or ezetimibe and bempedoic acid will be assessed.

7.2 Secondary Outcomes

- Plasma concentration of AZD0780.**
- Area under concentration-curve from time 0 to the last quantifiable concentration (AUClast).**
- Area under concentration-time curve in the dosing interval (AUCtau) (Day 15).**

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous evaluation via investigator interviews, laboratory assessments, and vital signs during the 28-day treatment period.

Serious Adverse Events (SAEs): Required to be reported to the Sponsor/Medical Monitor within 24 hours of site awareness.

Data Monitoring Committee (DMC): Internal safety review committee overseeing Phase 1 tolerance data.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for efficacy; Safety Set for all participants who received at least one dose of study

medication; PK Analysis Set for pharmacokinetic parameters.
Sample Size Justification: $N=81$ is deemed sufficient for a Phase 1 study to robustly evaluate safety signals, PK properties, and exploratory reductions in LDL-C. **Handling of Missing Data:** Standard Phase 1 methodology utilizing Mixed-effect Model for Repeated Measures (MMRM) for continuous endpoints.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: High-frequency onsite clinical monitoring typical of Phase 1 clinical research units. **Audit Trail:** eCRF entry within specified timeframes, electronic data capture, and systematic data clarification forms.

11. Study Identifiers & Governance

Primary ID: D7960C00017 **Registry Number:** NCT06742853 **Posting ID:** 869235349604422440 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** AZD0780 with Ezetimibe Combinations **Official Regulatory Title:** A Phase 1, Randomized, Single-blind, Placebo-controlled Study to Investigate the Pharmacokinetics, Safety, Tolerability, and Efficacy of AZD0780 in Combination With Ezetimibe, Ezetimibe/Rosuvastatin, or Ezetimibe/Bempedoic Acid in Healthy Male and Female Participants 18 to 75 Years of Age With Elevated LDL-C.

12. Clinical Framework (Dyslipidemia Specific)

Primary Condition / Disease Area: Dyslipidemia **Preferred UMLS Name:** Hypercholesterolemia **Diagnosis Threshold:** Fasting LDL-C ≥ 100 mg/dL but < 190 mg/dL **Medical Methodology:** Lipid-lowering combination therapy (PCSK9 inhibitor + standard of care) **Optimization Target:** Change in direct LDL-C from baseline

13. Study Procedures & Methodology

Management Pathway: Phase 1 safety, tolerability, and PK/PD assessment **Education Protocol (HCP):** Investigator training on protocol requirements and PK sampling **Education Protocol (Patient):** Informed consent, fasting requirements, and protocol adherence **Quality Audit Mechanism:** Clinical site monitoring, source data verification, and PK sample tracking

14. Observation & Follow-Up Schedule

Total Duration: Up to 14 weeks (including 4-week screening, 4-week run-in, 4-week treatment, and 2-week follow-up period)

Onsite Visit Cadence: Frequent visits during the 28-day treatment period for intensive PK/PD sampling, plus follow-up visits one and two weeks after the last dose of study drug. **Remote**

Monitoring Frequency: N/A (Standard Phase 1 onsite requirements apply) **Checklist Review Interval:** End of the 4-week treatment

period and 2-week follow-up phase.

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				